

AUTOMATIC BAR & STRUCTURAL ROLLING MILLS,
HORIZONTAL / VERTICAL STANDS
HOUSING-LESS STANDS
AUTOMATIC COOLING BEDS.

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INTRODUCTION

Steefo Engineering Corporation is an engineering company with the State of the art manufacturing facility, specialized in the field of Rolling Mill & Billet Plant technology and is fully committed and dedicated to the steel and related industries.

Supply of various equipment and process technology for Rolling Mill & Billet Plant are one of the core activities of Steefo Engineering and we design, manufacture and supply Turn Key and tailor-made systems for the Steel Plant.

In the last 45 years Steefo Engineering has acquired the technology and Know How for the entire production of quality steels by working with metallurgists and process experts who have worked and gained valuable experience in the metal production industry worldwide.

Steefo Engineering is able to design and built single units as well as whole production plants for the Rolling Mill & Billet Plant from one hand ensuring single cast design of the entire plant.

We have done Steel Plants of Reputed Groups on Turn Key basis in around 20 countries, including India, Indonesia, Saudi Arabia, Jordan, Qatar, Angola, D.R of Congo, Ethiopia, Mozambique, Kenya , Uganda; Zimbabwe, Mali, Nigeria, Ghana, Rwanda, Sri Lanka, Bangladesh, Gabon, Sudan, Benin, UAE and Mauritius.

All our above projects are working satisfactorily due to good quality of finish material, low cost of production, low maintenance and better after sales service.



GENERAL SPECIFICATIONS AND DESCRIPTION OF ROLLING MILL**1. TECHNICAL PARAMETERS**

Mill Production:	120,000 Tons / Year
Basic Raw Material:	Billets 100 to 130mm
Finished Products:	8mm Ø to 32 mm High Strength TMT Re Bars

2. PRE-CONDITIONS OF BILLETS**TOLERANCES**

Square Billets:	Side length ± 1.5 % max. Diagonal < 6 mm
Cut to Length:	± 50 mm
Bending:	Not more than 10mm in 1m Not more than 40mm in 12m Billet without distinct twist, irrectangular Billet angular twist shall not be greater than $1^{\circ}/m$

3. STEEL GRADE

Common low and medium carbon steel for rebars
according to standard BS 4449 Grade DIN 448

4. TOLERANCES OF FINISHED PRODUCTS

Finished products according to normal B.S. standard or equivalent to any international standard.

5. STARTING TEMPERATURE

Billet temperature in Roughing Stand: $1,100^{\circ}\text{C}$ approx.



A. BASIC DATA AND MEDIA REQUIREMENTS**1. PLANT POWER CONDITION**

- 1.1 AC - High Voltage: 30 KV $\pm$ 10% - 3 Ph (or as per Local available)
- Low Voltage: 400V, $\pm$ 10% 3 Ph
- Frequency: 50 Hz
- 1.2 - Solenoid Valves : 24V DC
- Limit Switches – Photocell: DC 24 V
- Control Circuit: DC 24V

2. CLEAN WATER SYSTEM (CLOSED CIRCUIT)

Water flow rate:
Mill Water : 500m³ / hr approx.
Operating pressure: approx. 4 bar for Mill
TMT Water : 500m³ / hr approx.
approx. 16 Bar for TMT
Temperature: max 35°C approx.

3. COMPRESSED AIR**Air Consumption**

- Free Air: 270 Nm³ / Hr

4. HYDRAULIC OIL

Initial filling oil for hydraulic 1000 litres (app.)
Central unit:
Enklo 68 (H.P.) / Equivalent

5. LUBRICATION OIL

Initial filling oil for lubrication
Unit Gear boxes etc. 4,000 Litres (app.)

Note: oil having the following properties will be used

- GEAR OIL VG - 32
- Pour point °C max: - 15
- Viscosity c at 40°C: 88.8
- Flash point °C: 246

6. LUBRICATION GREASE

Initial filling grease for central unit

- Lithon EP (IOC) or equivalent : 200 Kgs
- EP 350 (H.P.) : 100 Kgs
- High Temperature Moly Coat : 200 Kgs

7. ROLL CONSUMPTION: 0.4 to 0.6 Kg / Ton

8. POWER REQUIREMENT 3 MW



AUTOMATIC BAR ROLLING MILL**SUMMARY****A. MILL EQUIPMENT**

- | | |
|--|----------|
| 1. 510mm (20") Heavy Duty Roughing Mill | 1 Stand |
| 2. 410mm (16") Continuous Intermediate Mill | 2 Stands |
| 3. 360mm (14") Continuous Intermediate Mill | 4 Stands |
| 4. 320mm (12") Continuous Intermediate Mill | 4 Stands |
| 5. 280mm (11") Continuous Finishing Mill | 6 Stands |
| 6. Auxiliary Equipment. | |
| 7. Electricals | |
| 8. Automatic Cooling Bed with Twin Channel & Cold Shear | |
| 9. Continuous Operating Dividing Shear for TMT Bars | |
| 10. Bar Finishing, Bundling & Semi Automatic Tying Machine (Packing Bed) | |
| 11. TMT System | |
| 12. UTM Machine | |

B. Consultation & Supervision of Erection, Installation & Commissioning of Rolling Mill

A. Mill Equipment**1. 510mm Heavy Duty Roughing Mill****1.1 High Speed Flywheel****1 No**

Steel Flywheel running @ 740 rpm
app. 8 Tons weight complete with shaft,
spherical roller bearings, pedestals and base plate.

1.2 Reduction Gear Box.**2 Nos.**

Input - 740 rpm
Output - 95 rpm
Nominal Power - 1120 KW
Peak load - 2240 KW
Momentary Load - 3360 KW

Construction:

- Gears & Pinions will be of forged Steel, shrunk fitted on En-9 forged steel shafts.
- Helical teeth Profile on Gear Hobbing Machine
- Shafts end with key way.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Piping provided for forced lubrication arrangement of bearings and pinion teeth.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Pinion enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.

Transparent window provided on both sides of the housing to observe teeth and lubrication



1.3 510mm 3 -Hi Pinion Gear Box.**1 No.**

Pinion Centre - 510 mm
Nominal Power - 1120 KW
Peak load - 2240 KW
Momentary Load - 3360 KW

Construction:

- Gears & Pinions will be of forged Steel, shrunk fitted on En-9 forged steel shafts.
- Helical teeth Profile on Gear Hobbing Machine
- Shafts end with key way.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Piping provided for forced lubrication arrangement of bearings and pinion teeth.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Pinion enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.

Transparent window provided on both sides of the housing to observe teeth and lubrication

1.4 Base Plate for Pinion Gear Box**2 Nos.**

1.5 Mill Stand**1 No.**

Type : 3 Hi
Size : 510MM
Roll Dia. : Max. 550 MM
Min. 490 MM
Barrel : 1350 MM
Roll Neck Bearing : Spherical Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm² U.T.S.
Chocks are accurately machined and are inter changeable.
Top Roll Adjustment : By screw down mechanism.

- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
- The cap shall be fixed over the housings by means of two taper wedges which gives very good rigidity and does not loose during Mill operation.
- Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
- The accurate machining of parts and housings keep vibration level to minimum during rolling.

1.6 Cardan Shaft & Universal Head**3 Nos.**

Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.



- 1.7 Full Geared Heavy Duty Couplings** **4 Set**
Type : full flexible Geared Coupling to be connected between
Reduction Gear Box, Pinion Stand and Motor
- Design Parameters**
Normal Rating : 1120 kW
RPM : 750
Service Factor : ≥ 3
Type of Drive : Non reversible for all
Duty : Rolling Mill Duty
- 1.8 Foundation Bolts** **1 Set**
- 1.9 Rolls (1 Set)** **3 Nos.**
- 1.10 Roll Neck Bearing (1 Set)** **6 Nos.**
- 1.11 Motor Base Plate** **1 Set**



2. 410mm Continuous Intermediate Mill – 2 Stand**2.1 Reduction cum Pinion Gear Box****2 Nos.**

Normal Rating	:	300 KW
RPM	:	750-1500
Service Factor	:	2.5 (On Load)
Duty	:	Rolling Mill
Bearings	:	SPH. Roller Bearings

Construction:

- Gears & Pinions will be of forged 18CrNiMo-7/case Carburized steel hardened to 60±2 HRC, shrunk fitted on En-9 forged steel shafts. The gears and pinions will conform to DIN 3 / 4 Standard.
- Single helical teeth hardened and Profile Ground on Gear grinder.
- Shafts end with key way.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Piping provided for forced lubrication arrangement of bearings and pinion teeth.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Pinion enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.
- Transparent window provided on both sides of the housing to observe teeth and lubrication.



2.2 410 mm Mill Stands**2 Nos.**

Type : 2 Hi
Size : 410MM
Roll Dia. : Max. 450 MM
Min. 390 MM
Barrel : 800 MM
Roll Neck Bearing : Spherical Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm² U.T.S.

Chocks are accurately machined and are inter changeable.

Top Roll Adjustment : By screw down mechanism.

- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
- The cap shall be fixed over the housings by means of two tapper wedges which gives very good rigidity and does not loose during Mill operation.
- Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
- The accurate machining of parts and housings keep vibration level to minimum during rolling.

2.3 Cardan Shafts**4 Nos.**

Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.



- 2.4 Geared Coupling** **2 Nos.**
Type : full flexible Geared Coupling to be connected between
Reduction Gear Box, Pinion Stand and Motor
- Design Parameters**
Normal Rating : 300 kW
RPM : 750 - 1500
Service Factor : ≥ 2.5
Type of Drive : Non reversible for all
Duty : Rolling Mill Duty
- 2.5 Foundation Bolts** **1 Set**
- 2.6 Rolls (1 Set)** **4 Nos.**
- 2.7 Roll Neck Bearings (1 Set)** **8 Nos.**
- 2.8 Motor Base Plate** **8 Nos.**



3. 360mm Continuous Intermediate Mill – 4 Stand**3.1 Reduction cum Pinion Gear Box****4 Nos.**

Normal Rating	:	300 KW
RPM	:	750-1500
Service Factor	:	2.5 (On Load)
Duty	:	Rolling Mill
Bearings	:	SPH. Roller Bearings

Construction:

- Gears & Pinions will be of forged 18CrNiMo-7/case Carburized steel hardened to 60±2 HRC, shrunk fitted on En-9 forged steel shafts. The gears and pinions will conform to DIN 3 / 4 Standard.
- Single helical teeth hardened and Profile Ground on Gear grinder.
- Shafts end with key way.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Piping provided for forced lubrication arrangement of bearings and pinion teeth.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Pinion enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.
- Transparent window provided on both sides of the housing to observe teeth and lubrication.



3.2 360 mm Mill Stands**4 Nos.**

- Type : 2 Hi
Size : 360MM
Roll Dia. : Max. 400 MM
Min. 340 MM
Barrel : 700 MM
Roll Neck Bearing : Spherical Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm² U.T.S.
Chocks are accurately machined and are inter changeable.
Top Roll Adjustment : By screw down mechanism.
- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
 - The cap shall be fixed over the housings by means of two taper wedges which gives very good rigidity and does not loose during Mill operation.
 - Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
 - The accurate machining of parts and housings keep vibration level to minimum during rolling.

3.3 Cardan Shafts**8 Nos.**

- Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.



- 3.4 Geared Coupling** **4 Nos.**
Type : full flexible Geared Coupling to be connected between
Reduction Gear Box, Pinion Stand and Motor
- Design Parameters**
Normal Rating : 300 kW
RPM : 750 - 1500
Service Factor : ≥ 2.5
Type of Drive : Non reversible for all
Duty : Rolling Mill Duty
- 3.5 Foundation Bolts** **1 Set**
- 3.6 Rolls (1 Set)** **8 Nos.**
- 3.7 Roll Neck Bearings (1 Set)** **16 Nos.**
- 3.8 Motor Base Plate** **16 Nos.**



4. 320mm Continuous Intermediate Mill – 4 Stand**4.1 Reduction cum Pinion Gear Box****4 Nos.**

Normal Rating	:	300 KW
RPM	:	1000-2000
Service Factor	:	2.5 (On Load)
Duty	:	Rolling Mill
Bearings	:	SPH. Roller Bearings

Construction:

- Gears & Pinions will be of forged 18CrNiMo-7/case Carburized steel hardened to 60±2 HRC, shrunk fitted on En-9 forged steel shafts. The gears and pinions will conform to DIN 3 / 4 Standard.
- Single helical teeth hardened and Profile Ground on Gear grinder.
- Shafts end with key way.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Piping provided for forced lubrication arrangement of bearings and pinion teeth.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Pinion enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.
- Transparent window provided on both sides of the housing to observe teeth and lubrication.



4.2 320 mm Mill Stands**4 Nos.**

- Type : 2 Hi
Size : 320MM
Roll Dia. : Max. 360 MM
Min. 300 MM
Barrel : 600 MM
Roll Neck Bearing : Spherical Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm² U.T.S.
Chocks are accurately machined and are inter changeable.
Top Roll Adjustment : By screw down mechanism.
- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
 - The cap shall be fixed over the housings by means of two taper wedges which gives very good rigidity and does not loose during Mill operation.
 - Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
 - The accurate machining of parts and housings keep vibration level to minimum during rolling.

4.3 Cardan Shafts**8 Nos.**

- Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.



- 4.4 Geared Coupling** **4 Nos.**
Type : full flexible Geared Coupling to be connected between
Reduction Gear Box, Pinion Stand and Motor
- Design Parameters**
Normal Rating : 300 kW
RPM : 1000 - 2000
Service Factor : ≥ 2.5
Type of Drive : Non reversible for all
Duty : Rolling Mill Duty
- 4.5 Foundation Bolts** **1 Set**
- 4.6 Rolls (1 Set)** **8 Nos.**
- 4.7 Roll Neck Bearings (1 Set)** **16 Nos.**
- 4.8 Motor Base Plate** **16 Nos.**



5. 280mm Continuous Finishing Mill – 6 Stand**5.1 Reduction cum Pinion Gear Box****6 Nos.**

Normal Rating	:	300 KW
RPM	:	1000-2000
Service Factor	:	2.5 (On Load)
Duty	:	Rolling Mill
Bearings	:	SPH. Roller Bearings

Construction:

- Gears & Pinions will be of forged 18CrNiMo-7/case Carburized steel hardened to 60±2 HRC, shrunk fitted on En-9 forged steel shafts. The gears and pinions will conform to DIN 3 / 4 Standard.
- Single helical teeth hardened and Profile Ground on Gear grinder.
- Shafts end with key way.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Piping provided for forced lubrication arrangement of bearings and pinion teeth.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Pinion enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.
- Transparent window provided on both sides of the housing to observe teeth and lubrication.



5.2 280mm Mill Stands**6 Nos.**

Type	: 2 Hi
Size	: 280MM
Roll Dia.	: Max. 310 MM Min. 260 MM
Barrel	: 500 MM
Roll Neck Bearing	: Roller Bearing
Housing & Cap	: Cast Steel – Grade-2 (IS-1030)
All Fasteners	: High Tensile Steel Gr.8.8 min.
Material for Chocks	: Cast Steel with 500-600 N/mm ² U.T.S.

Chocks are accurately machined and are inter changeable.

Top Roll Adjustment : By screw down mechanism.

- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
- The cap shall be fixed over the housings by means of two taper wedges which gives very good rigidity and does not loose during Mill operation.
- Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
- The accurate machining of parts and housings keep vibration level to minimum during rolling.

5.3 Cardan Shafts**12 Nos.**

Location : Between Pinion Stand and rolls

Type : Cross Joint Type

Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.



- 5.4 Geared Coupling** **6 Nos.**
Type : full flexible Geared Coupling to be connected between
Reduction Gear Box, Pinion Stand and Motor
- Design Parameters**
Normal Rating : 300 kW
RPM : 1000 - 2000
Service Factor : ≥ 2.5
Type of Drive : Non reversible for all
Duty : Rolling Mill Duty
- 5.5 Foundation Bolts** **1 Set**
- 5.6 Rolls (1 Set)** **12 Nos.**
- 5.7 Roll Neck Bearings (1 Set)** **24 Nos.**
- 5.8 Motor Base Plate** **24 Nos.**



6. Auxiliary Equipment.**6.1 Y Table for Roughing Mill to feed Top passes automatic**

Length : 10 M (approx.)

Roller Barrel Length: 1200 mm

Roller Spacing : app. 800 mm

Frame : Fabricated from Channels

Drive- Electro- mechanical Group Driven with Chain and Sprocket Linear Speed-
App. 3 m/sec,

Roller Neck Bearings : Antifriction Spherical Roller Bearing

6.2 Fix Roller Table for Roughing Mill to feed Bottom passes automatic

Roller Diameter: 273 mm Low alloy steel seamless pipe

Roller Barrel Length: 1200 mm

Roller Spacing : app. 800 mm

Frame : Fabricated from Channels

Drive- Electro- mechanical Group Driven with Chain and Sprocket Linear Speed-
App. 3 m/sec,

Roller Neck Bearings : Antifriction Spherical Roller Bearing

6.3 Approach Roller Table to 1st Continuous Mill (Extension Roller Table)

Length -15 m (App.)

Roller Diameter- 246 mm low alloy steel seamless pipe

Barrel length- 700 mm

Drive : Group driven with chain & sprocket – 2 Nos.

With Electric Motor

Roll Neck Bearings : Anti Friction Pillow Blocks

6.4 Pinch roll Before Quenching Box

Nominal diameter- 300 mm

Barrel length- 100 mm

Motor : DC Motor with drive

Drive : With reduction gearbox and carbon shaft

Pneumatic by operation of Top Roll

Lubrication : Splash Oil

Drive Coupling : Flexible Gear Coupling

Gears & Shaft Material : Heat Treatment Steel

Entry & Exit Guides : Steel Fabricated



6.5 Pinch roll after Quenching Box

Nominal diameter- 300 mm
Barrel length- 100 mm
Motor : DC Motor with drive
Drive : with reduction gearbox and cardon shaft
Pneumatic by operation of Top Roll
Lubrication : Splash Oil
Drive Coupling : Flexible Gear Coupling
Gears & Shaft Material : Heat Treatment Steel
Entry & Exit Guides : Steel Fabricated

6.6 Dividing Shear for, Dia 8mm to Dia 32 mm (TMT)

Type : Direct Driven Dividing Shear (Steel Fabricated)
TMT Bar to be Cut : 10mm to 32mm
Bar Speed : Max. 16 m/s
Motor : DC Motor with drive & Base Plate
Shear with Lubrication Piping, Oil Supply System, Bar Shifter,
Delivery Guide Trough & Valve Stand
Gears : Case Hardened & Ground with Surface Hardness HRC 58-60
Shafts : Heat Treated Steel
Blades : Heat Treated Steel (46-50 HRC)

6.7 Start/ Stop type Crop & Cobble Shear – 2 Nos.

For Hot Material Cutting (Temp. 850°C to 1050°C)
Type : Direct Driven Dividing Shear (Steel Fabricated)
Motor : DC Motor with drive & Base Plate
Low Speed Start / Stop Shear with Lubrication Piping, Oil Supply System,
Delivery Guide Trough
Gears : Case Hardened & Ground with Surface Hardness HRC 58-60
Shafts : Heat Treated Steel
Blades : Heat Treated Steel (46-50 HRC)
shall be installed in between stand # C-2 & C-3.

6.8 Tail Breaking Pinch Rolls – 2 Nos.

Nominal diameter- 300 mm
Barrel length- 100 mm
Motor : DC Motor with drive
Drive : With reduction gearbox and cardon shaft
Pneumatic by operation of Top Roll
Lubrication : Splash Oil
Drive Coupling : Flexible Gear Coupling
Gears & Shaft Material : Heat Treatment Steel
Entry & Exit Guides : Steel Fabricated



6.9 Lubrication Systems – 4 Nos.

- Duplex Oil Lubrication System – 3 Nos.
- Technical Specifications:
 - a. Reservoir 4500 Ltr - 1 No.
 - b. Oil Pump 300 Lpm - 1 No.
 - c. Electric Motor 10 Hp - 2 Nos.
(1 No. Working & 1 No. Stand By)
- For above:
 - a. Oil Filter Duplex Micron 149 - 1 Set
 - b. Oil Cooler 30000K Cal - 1 No.
 - c.
- Instruments:
 - a. Pressure Gauge- 2 Nos.
 - b. Pressure Switches- 1 No.
 - c. Temperature Gauges- 2 Nos.
 - d. Diff Pressure Switch- 1 No.
 - e. Level Switch- 1 No.
 - f. Level Gauge- 1 Set
- Oil Lubrication System for Continuous Shear – 1 No.

6.10 Roller Guide Box with Holder 7 Nos.**6.11 Roller Twist Guide 16 Nos.****6.12 Funnels 1 Set****6.13 Static Guides 1 Set****6.14 Water pump 2 Nos.**

7. ELECTRICALS

1.	AC Motor 1120 KW for Roughing Mill	1 No.
2.	D.C. Motor for 410mm Continuous Mill	2 Nos.
3.	D.C. Motor for 360mm Continuous Mill	4 Nos.
4.	D.C. Motor for 320mm Continuous Mill	4 Nos.
5.	D.C. Motor for 280mm Continuous Mill	6 Nos.
6.	Small AC Motors for Y Table, Turning Wall Roller table, Approach Roller table, etc.	1 Set
7.	Main LT A.C. Incomer Panel for all AC / DC Motors	1 Set
8.	D.C Drive Control Panel with Control Desk for Shears & Pinch Rolls	8 Nos.
9.	Scada base DC Drives Control Panel with Control Desk for 410mm, 360mm, 320mm & 280mm Continuous Mill	14 Nos.
10.	PLC, PCs, Remote I/Os etc System for Rolling Mill	1 Set
11.	Motorized Liquid Resistance starter for 1120 KW Motor	1 No.
12.	DC Motors for Shears & Pinch Roll	8 Nos.

Note : HT Side, Transformer, all Cables & Capacitors & Earthing's are not included



8. Automatic Cooling Bed with Twin Channel & Cold Shear
Size 66 Mtr.**a. Twin Channel**

Application : Delivery of Rolled and cut Bars on to the Cooling Bed
Type : Electric Motor Driven single shaft Twin Channel
Bar Size : Dia 8mm – 32mm
Length : 66 Mtrs.
Maximum Speed : 20 M/Sec
Channels : S.G Iron Casting 1.1m long each
Central Box : 2.2m long Cast Iron, Air Cooled
Shafts : 2.2m Meter Long steel shaft
Bearings : Supported by pillow block
Cam : Forged Steel
Cam Followers: Roller Bearing Mounted on Pins in the C Channel
Coupling : Flexible
Drive : DC Electric Motor for line shaft operation

The Twin Channel bar delivery equipment consists of the following

- C type S.G. Iron Channels on both sides butting against cast Iron Central Box and hinged on top of the Box
- Box suspended through lugs from longitudinal frame on top
- Cam follower rollers mounted on a lug on each channel
- Single shaft is mounted on bearings and suspended from the longitudinal frame
- Cams for each C Channel are mounted on shaft
- Electric motor drive to main single shaft for movement

Operation

Both the C Channel (Right & Left) are closed. Rolled Re bar enters one of the C Channel and is cut to Cooling bed length while travelling by dividing shear. After cut, the bar tail end is braked using the tail braking pinch roll, so that the bar stops before its front end can shoot out of the channel. Motor operated cam shaft opens the C Channel to drop the bar on the Cooling bed straightening pockets. After dropping the bar, the channel closes immediately through rotation and is ready for the next bar. As soon the bar is cut, the next part of the bar is directed toward the other channel, which is ready to receive the same.



b. AUTOMATIC RAKE TYPE COOLING BED

Type	: Rake Type
Size	: 66 m long
Bar speed	: 22 m/sec (max)
In-let temperature of bar	: 600°C
Out-let temperature of bar	: 250°C

The cooling bed proposed is of automatic rake type and has the following:-

- i) Fixed Grid
- ii) Moving rakes
- iii) Eccentrics and drives.
- iv) Fixed rakes
- v) Aligning roller

i) FIXED GRID

Grid length	: 66 m
Notch pitch	: 66 mm
Grid material	: cast iron

The fixed grid is a simple cast iron construction with notches profiled identical to rake notches. They receive bars from diverter aprons. The moving rakes picks up the material from the fixed grid. Recess is provided for the movement of rakes. The fixed grid is provided in suitably sized segments.

ii) MOVING RAKES

Rake Length	: To fit size
Rake Pitch	: 600 mm
Notch Pitch	: 66 mm
Rake Material	: MS flats
Rake Mounting	: On MS fabricated frame
Rake Material	: MS flats
Rake Mounting	: On MS fabricated frame
Rake Thickness	: 16 mm

The moving rakes transports the bars towards the discharge end moving the material by one notch for every cycle, during the period cooling the bar by natural currents. The movement of the rakes is through the eccentrics.



iii) ECCENTRICS & DRIVES

Rake Drive : Through eccentrics

Eccentricity : 36 mm

Line Shaft Bearings : Spherical roller bearings housed in standard Plummer Blocks

Line Shaft Material: EN8

Eccentric Material : EN9

Line Shaft Inter Couplings: Muff type

Couplings between Line shaft and Worm Reduction Gearbox: Half Gear Coupling No. 108

Couplings between Worm Reduction Gearboxes & Reduction Gearbox and Motor: Half Gear Coupling No. 105

Drive mechanical : Worm Gear Boxes

The eccentrics are mounted on line shafts, housed in bearing housing bolted to moving rake mounting frame. The line shafts are coupled to gearboxes driven by a DC motor. The arrangement provides for one rotation of eccentric per cycle, which in effect moves the material by bone rake pitch.

iv) FIXED RAKES

Rake Length : To fit size

Rake Pitch : 300 mm

Notch Pitch : 66 mm

Rake Material : MS flats

Rake Mounting : On beams anchored to foundation.

The fixed rakes hold the bars during the rest of the periods.

v) ALIGNMENT ROLLS

Roll Diameter : ~200 mm

Bearings : Anti friction

Roll Lengths : ~ 600 mm

Notch Profile : To suit rake

Roll Drive : Geared motor

Towards the discharge end are placed a series of aligning rolls which aligns the bar front ends to facilitate cutting at cold shear. These are cast iron driven rolls profiled identical to the rakes and positioned between rakes.



c. Run Out Roller Table 1 Set

Run- out roller table

- a) Roller Pitch – 1500mm
- b) Roller size 216 Ø x 500 B.L.
- c) Spherical roller bearing
- d) MS Fabrication Table Frame
- e) Length 66 Mtr.

d. Cold Shear 1 No.

Capacity : 300 Ton
Blade width : 600 mm
Nos. of cuts : 8 cuts/ min
Construction : Fabricated steel body, heat-treated steel gears,
shafts & blades, shafts mounted on roller bearings
Drives : AC Motor

9 Continuous operating dividing shear for TMT Bars

Type : Direct Driven Dividing Shear (Steel Fabricated)

Bar Speed : Max. 20 m/s

High speeds Continuous Shear with Lubrication Piping, Oil Supply System, Servo Bar Shifter

Delivery Guide Trough & Valve Stand

Gears : Helical, Case Hardened & Ground with Surface Hardness HRC 58-60

Shafts : Heat Treated Steel

Blades : Heat Treated Steel (46-50 HRC)



10. Bar Finishing, Bundling (Packing Bed)**10.1 Cold Shear Run – Out Roller Table (Gauge Conveyor)**

Purpose : Convey the material to bar transfer run-in roller table

Total Length : 26000mm

Table Length (C/C to end rollers) : 24000mm

Roller Spacing : 1200mm

Rollers : Diameter 190mm,

Barrel Length – 800mm,

Speed : 1.9 m/sec

Frame : Steel Fabricated

Drive : Group Driven with chain & sprocket through reduction gear box

Roller Neck Bearings : Pillow Blocks

Disappearing Stopper : 2 no. of pneumatically operated disappearing stoppers shall be installed at the exit of cold shear run-out roller table to stop the bar at 6/12 m from cold shear

10.2 Packing Bed

1) Size: 12 MW x 7.0 ML

2) Transfer Inlet-Arm unit x 12 M:

A) Main Shaft: Ø75 x 12 ML

B) Main drive motor: AC-10 HP x 4P x 1/10

C) Main Wrom Gear Reducer: A-200 x 1/30

D) Gear coupling: TY –SB- 150 x 2

3) Conveyor chain

4) Sprocket

5) Conveyor chain driven unit x 12ML

A) Main Shaft: Ø65 x 12ML

B) Main drive motor: AC –10HP x 4P x 1/10

C) Main Wrom Gear Reducer

D) Gear Coupling



10.3 Semi Automatic Tying Machine - 2 Nos.

- Tying Machines are for semi automatic tying of round bundles of bar.
- The Machines are fitted on a frame which is located in the line of roller table and the position is fixed.
- The tying material is short bars has been cut to 600mm in length and are fed through the machine manually.
- The Tying unit makes a twisted circle knot by means of a special head driven by an electrical motor and it is controlled by limit switches and cylinder moving up and down.
- The bundles are well tightened by means of two special arms making it almost circular.
- Electrical panel and PLC unit located on the pulpit is also in the scope of supply

Bundle dimensions	: Dia 200 – 400mm
Bundle Weight	: 2 Ton
Tying Material	: Short bars Dia 8 to 12mm
Bundle Forming Device	: Rotating Arms
Twisted Head	: 1 No. Electromechanically Drive
Tying Period (5 Ties)	: 85 sec
Tip Laying Rolls	: 1 No.



10.4 Roller Conveyor

- 1) Total Length = 19,000 (mm)
- 2) V-roller before bundling machine = 5,000 (mm)
 - 2-1) Conveyor drive unit:
 - i. Drive gear motor: AC-7.5 HP x 4P x 1/19
 - ii. Conveyor roller: Ø216.3 x 450W
 - iii. Coupling: RF-265
 - iv. Roller Chain & Sprocket: RS 100
 - 2-2) Free V-roller unit:
 - a) V-roller: Ø 89.1 x 406W
 - b) Bearing: 6207Z
 - 2-3) Bundle are drive unit:
 - a) Driver gear motor: AC-5HP x 4P x 1/19
 - b) Conveyor roller: Ø216.3 x 450W
 - c) Roller Chain & Sprocket: RS100
 - 2-4) Clamp by air cylinder: TC- Ø250 x 400 ST
- 3) Roller Conveyor after bundling = 14,000 (mm)
 - 3-1) Conveyor drive unit:
 - v. Driver gear motor: AC-7.5 HP x 4P
 - vi. Conveyor roller: Ø267.4 x 600W
 - 3-2) Clamp by air cylinder: TC- Ø250 x 400 ST
 - 3-3) Stopper: TC- Ø250 x 530ST



11. TMT System**TECHNICAL PARAMETERS**

- | | |
|---------------------------------|--|
| 1. Capacity | 120,000 Tons / Year |
| 2. Mill | Bar Mill |
| 3. Input Size | Billets 100 to 130mm |
| 4. Sizes to be processed | 8 - 32 mm. |
| 5. Temperature after last stand | Max. 1000°C for 8 - 12mm dia.
Max. 1050°C for 16 - 32mm dia. |
| 6. Temperature difference | Less than 50°C between head and tail ends
after last stand |
| 7. Chemical Composition | As per IS 1786 : 2008 |
| 8. Aim | Production of High strength rebars as per
IS 1786 : 2008
Grades Fe 415, Fe 415D, Fe 500, Fe 550D,
Fe 550, Fe 550D and Fe 600. |



TECHNICAL DESCRIPTION

1 Introduction

For the purpose of controlled processing of deformed steel bars by means of high pressure water, Rolling Mill TMT System will be installed in the Rolling Mill with the aim to produce 'Quenched and Self-Tempered' (QST) steel rebars. A suitable shear needs to be located after the Rolling Mill TMT System to cut the rebars to suit the Cooling Bed length.

2. Technical parameters

The bars produced through the Rolling Mill TMT System as per this offer shall produce rebars as per the IS 1786: 2008 (Fourth Revision) amendment 3 of March 2017 Grades Fe 415, Fe 415D, Fe 500, Fe 500D, Fe 550, Fe 550D and Fe 600.

3. Layout

3.1 Main Rolling Mill TMT system with Pinch Roll Unit.

The main TMT system consists of the Rolling Mill TMT Pipes installed after the finishing rolling stand but before the cooling bed. Two pinch roll units need to be located before and after the Rolling Mill TMT pipes.

3.1.1 Arrangement of Rolling Mill TMT pipes

For dimension range of Ø10 to 32 mm, adequate number and type of Rolling Mill TMT pipes of different lengths will be used. The length of the Rolling Mill TMT Pipes would suit mill parameters.

Fabricated base frame trolley(s) for mounting of the desired Rolling Mill TMT pipes will be provided for the system.

In the TMT Line, Pinch roll units with speed control are needed - to ensure that the tail end of rebar is thermo-processed in the same way as the front and middle portion of the rebar to obtain uniform properties over the entire length.

3.1.2 Water

High pressure water pumps will be provided so that water at desired pressure is available for rapid water quenching of rebars to obtaining desired properties.

3.1.3 Compressed air consumption

- Air Compressor

About 250 CFM at 7 bar will be needed to be provided by client.

3.1.4 Pressure water pump station

The pump station has high-pressure water pumps, including one standby pump, which will deliver water into a main pipe.

A motor operated valve controls water flow.



3.1.5 Details of the water and air-piping scheme

The Rolling Mill TMT pipes are fed from the pressure water manifold through flexible hoses that are controlled by manually operated hand valves. Each rolling mill TMT pipe has two water inlets.

The valves have only two positions "closed" and "open". This makes the operation very simple for the operators.

At the opposite end of the water inlets is arranged the air inlet. A manually operated valve controls the airflow.

3.1.6 Measuring and control instrumentation**3.1.6.1 Control instruments**

Located in a control cabin.

3.1.6.2 Pressure transmitter

A pressure transmitter with indicator and 4 - 20 mA signal mounted on water header near TMT system. The water pressure would be displayed on the digital meter located in the control cabin.

3.1.6.3 Temperatures

The final rolling temperature TF (measuring range 300 - 1300°C) is measured ahead of the TMT pipe.

The equalizing temperature TE (measuring range 300 - 1300°C) is measured on the cooling bed. For quality control purposes, the temperatures are indicated in the control cabin.

3.1.6.4 Check of tensile properties

At the beginning of a rolling campaign the tensile properties should be checked on samples as soon as possible and related to the equalizing temperature TE. This test is made by means of a tensile testing machine.

3.2 Cooling Tower & Scale Pit

The water used for water quenching as per the TMT Technology is collected in a scale pit via water flumes. It is then cooled at a Cooling Tower for re-use in the Rolling mill TMT system.

The hot water pumps and the Cooling Tower are included from our Scope of Supplies. Additionally, we shall provide the detail engineering for the water flumes and scale pit to enable the customer to build them as per local standards and practices.

3.3 Electrics & Controls

The electrics will operate from the supply voltage of 400V AC $\pm 10\%$, 50 Hz $\pm 5\%$.



SCOPE OF SUPPLY

The quantity and technical data given in this Chapter is indicative and subject to change. These will be finalized during detail engineering to ensure that the client is able to produce TMT rebars of international standard.

1 Main Rolling Mill TMT system

- 1 lot Rolling Mill TMT pipes with appropriate number of intermediate lengths.
- 1 lot manual valves PN 21 for water
- 1 lot flexible hoses PN 16 for water
- 1 lot manual valves R 1" PN 6 for compressed air
- 1 lot flexible hoses DN 20 PN 6 for compressed air
- 1 lot analogue pressure gauge display 0-25 Kg/cm² for water
- 1 lot motor controlled valve PN 21

2 Support Trolley for Rolling Mill TMT Pipes

The support base frame before cooling bed is to accommodate the different Rolling Mill TMT pipe lengths with holes bored to suit for mounting. When using short pipes, the remaining will be bridged with cast iron funnels

3 Piping

The interconnected pipes, with flanges and fittings will be supplied by us.

4 Temperature measuring instruments

- 1 No. Optical pyrometer, range 300 - 1300°C located immediately after the finishing stand;
- 1 No. Optical pyrometer, range 300 - 1300°C located in the centre of the cooling bed;

5 Pressure Transmitter

- 1 No. Pressure transmitter range 0 - 35 Kg/cm² located on the water header near TMT System.

6 Electrical equipment

The electrical equipment will be suitable to operate from 400 VAC $\pm$ 10%, 50 Hz $\pm$ 5% at 40°C ambient.

7 Controls

Controls would be located in the control room and pulpit and would be suitable to operate at an ambient temperature of 30°C.

8 High Pressure Pumps

Lot of high pressure pumps (including 1 standby) for pumping water from Cooling Tower basin to Rolling Mill TMT System will be supplied by us.

9 Cooling Tower

A Cooling Tower of adequate rating will be supplied by us.

10 Hot Water Pumps

Lot of hot water pumps (including 1 standby) for pumping water from the scale pit to the Cooling Tower will be supplied by us.



12. UTM Machine

12.1 Electronic Universal Testing Machine 1 No.

With PC Interface & Real Time Graph

(Panel / PC Controlled)

Capacity : 100 Ton

- Loading accuracy as high $\pm 1\%$.
- Straining at variable speeds to suit a wide range of materials.
- Printer & PC graphs enable study the behavior of the Material.
- Motor driven threaded columns for quick effortless adjustment of middle crosshead to facilitate rapid fixing of Test specimen.
- Simplicity in reading because of Digital readouts.
- Wide range of standard and special accessories including load stabilizer.
- Easy change from plain to threaded and screwed specimen.
- Large effective clearance between columns enable testing of standard specimens as well as structure.
- Simple Controls for ease of operation.
- Robust straining frame of an extremely rigid construction.
- Safe operation ensured by means of safety devices.
- Fully enclosed and protected pressure transducer.
- RS 232 serial port to transfer data to computer for analysis/ storage evaluation etc.
- Manual control & release valve operation.

12.2 Bend / Re Bend Machine – 1 No.



							S	STEEFO		
							C	CUSTOMER		
SCOPE OF SUPPLY OF AUTOMATIC ROLLING MILL										
Sr. No.	Description	Qty.	Basic Data	Basic Design	Detailed Design	Supply	Supervision of Erection	Erection & Comm.	Supervision of Comm.	
A	MILL EQUIPMENT									
1	510 MM Heavy Duty Roughing Mill									
1.1	High Speed Flywheel	1 No.	S	S	S	S	S	C	S	
1.2	Reduction Gear Box	2 Nos.	S	S	S	S	S	C	S	
1.3	510mm 3 -Hi Pinion Gear Box	1 No.	S	S	S	S	S	C	S	
1.4	Base Plate for Pinion Gear Box	2 Nos.	S	S	S	S	S	C	S	
1.5	Mill Stand	1 No.	S	S	S	S	S	C	S	
1.6	Cardan Shaft with Universal Head	3 Nos.	S	S	S	S	S	C	S	
1.7	Full Geared Heavy Duty Couplings	4 Set	S	S	S	S	S	C	S	
1.8	Foundation Bolts	1 Set	S	S	S	S	S	C	S	
1.9	Rolls (1 Set)	3 Nos.	S	S	S	S	S	C	S	
1.10	Roll Neck Bearings	6 Nos.	S	S	S	S	S	C	S	
1.11	Motor Base Plate	1 Set	S	S	S	S	S	C	S	
2	410mm Continuous Intermediate Mill									
2.1	Reduction cum Pinion Gear Box	2 Nos.	S	S	S	S	S	C	S	
2.2	410 mm Mill Stands	2 Nos.	S	S	S	S	S	C	S	
2.3	Cardan Shafts	4 Nos.	S	S	S	S	S	C	S	
2.4	Geared Coupling	2 Nos.	S	S	S	S	S	C	S	
2.5	Foundation Bolts	1 Set	S	S	S	S	S	C	S	
2.6	Rolls	4 Nos.	S	S	S	S	S	C	S	
2.7	Roll Neck Bearings	8 Nos.	S	S	S	S	S	C	S	
2.8	Motor Base Plate	8 Nos.	S	S	S	S	S	C	S	
3	360mm Continuous Intermediate Mill									
3.1	Reduction cum Pinion Gear Box	4 Nos.	S	S	S	S	S	C	S	
3.2	360 mm Mill Stands	4 Nos.	S	S	S	S	S	C	S	
3.3	Cardan Shafts	8 Nos.	S	S	S	S	S	C	S	
3.4	Geared Coupling	4 Nos.	S	S	S	S	S	C	S	
3.5	Foundation Bolts	1 Set	S	S	S	S	S	C	S	
3.6	Rolls	8 Nos.	S	S	S	S	S	C	S	
3.7	Roll Neck Bearings	16 Nos.	S	S	S	S	S	C	S	
3.8	Motor Base Plate	16 Nos.	S	S	S	S	S	C	S	



							S	STEEFO		
							C	CUSTOMER		
SCOPE OF SUPPLY OF AUTOMATIC ROLLING MILL										
Sr. No.	Description	Qty.	Basic Data	Basic Design	Detailed Design	Supply	Supervision of Erection	Erection & Comm.	Supervision of Comm.	
4	320mm Continuous Intermediate Mill									
4.1	Reduction cum Pinion Gear Box	4 Nos.	S	S	S	S	S	C	S	
4.2	320 mm Mill Stands	4 Nos.	S	S	S	S	S	C	S	
4.3	Cardan Shafts	8 Nos.	S	S	S	S	S	C	S	
4.4	Geared Coupling	4 Nos.	S	S	S	S	S	C	S	
4.5	Foundation Bolts	1 Set	S	S	S	S	S	C	S	
4.6	Rolls	8 Nos.	S	S	S	S	S	C	S	
4.7	Roll Neck Bearings	16 Nos.	S	S	S	S	S	C	S	
4.8	Motor Base Plate	16 Nos.	S	S	S	S	S	C	S	
5	280mm Continuous Finishing Mill.									
5.1	Reduction cum Pinion Gear Box	6 Nos.	S	S	S	S	S	C	S	
5.2	280 mm Mill Stands	6 Nos.	S	S	S	S	S	C	S	
5.3	Cardan Shafts	12 Nos	S	S	S	S	S	C	S	
5.4	Geared Coupling	6 Nos.	S	S	S	S	S	C	S	
5.5	Foundation Bolts	1 Set	S	S	S	S	S	C	S	
5.6	Rolls	12 Nos.	S	S	S	S	S	C	S	
5.7	Roll Neck Bearings	24 Nos.	S	S	S	S	S	C	S	
5.8	Motor Base Plate	24 Nos.	S	S	S	S	S	C	S	
6	Auxiliary Equipment									
6.1	Y Table for Roughing Mill to feed Top passes automatic	1 No.	S	S	S	S	S	C	S	
6.2	Fix Roller Table for Roughing Mill to feed Bottom passes automatic	1 No.	S	S	S	S	S	C	S	
6.3	Approach Roller Table to 1st Continuous Mill (Extension Roller Table)	1 No.	S	S	S	S	S	C	S	
6.4	Pinch roll Before Quenching Box	1 No.	S	S	S	S	S	C	S	
6.5	Pinch roll after Quenching Box	1 No.	S	S	S	S	S	C	S	
6.6	Dividing Shear for, Dia 10 mm to Dia 32 mm (TMT)	2 Nos.	S	S	S	S	S	C	S	
6.7	Start/ Stop type Crop & Cobble Shear	2 Nos.	S	S	S	S	S	C	S	
6.8	Tail Breaking Pinch Rolls	2 Nos.	S	S	S	S	S	C	S	
6.9	Lubrication Systems (5 Nos.)	1 Set	S	S	S	S	S	C	S	
6.10	Roller Guide Box with Holder	7 Nos.	S	S	S	S	S	C	S	
6.11	Roller Twist Guide	16 Nos.	S	S	S	S	S	C	S	
6.12	Funnels	1 Set	S	S	S	S	S	C	S	
6.13	Static Guides	1 Set	S	S	S	S	S	C	S	
6.14	Water pump	2 Nos.	S	S	S	S	S	C	S	



Sr. No.	Description	Qty.	Basic Data	Basic Design	Detailed Design	Supply	Supervision of Erection	Erection & Comm.	Supervision of Comm.
7	ELECTRICALS								
7.1	A.C. Slip ring Induction Motor 1120 KW / 740 RPM for Roughing Mill	1 No.	S	S	S	S	S	C	S
7.2	D.C. Motor for 410mm Continuous Mill	2 Nos.	S	S	S	S	S	C	S
7.3	D.C. Motor for 360mm Continuous Mill	4 Nos.	S	S	S	S	S	C	S
7.4	D.C. Motor for 320mm Continuous Mill	4 Nos.	S	S	S	S	S	C	S
7.5	D.C. Motor for 280mm Continuous Mill	6 Nos.	S	S	S	S	S	C	S
7.6	Small AC Motors for Y Table, Turning Wall Roller table, Approach Roller table, etc.	1 Set	S	S	S	S	S	C	S
7.7	Main LT A.C. Incomer Panel for all AC / DC Motors	1 Set	S	S	S	S	S	C	S
7.8	D.C Drive Control Panel with Control Desk for Shears & Pinch Rolls	8 Nos.	S	S	S	S	S	C	S
7.9	Scada base DC Drives Control Panel with Control Desk for 410mm, 360mm, 320mm & 280mm Continuous Mill	16 Nos.	S	S	S	S	S	C	S
7.10	PLC, PCs, Remote I/Os etc System for Rolling Mill	1 Set	S	S	S	S	S	C	S
7.11	Motorised Liquid Resistance starter for 1120 KW Motor	1 No.	S	S	S	S	S	C	S
7.12	DC Motors for Shears & Pinch Roll	8 Nos.	S	S	S	S	S	C	S
8	AUTOMATIC COOLING BED WITH TWIN CHANNEL & COLD SHEAR								
8.1	Twin Channel	1 Set	S	S	S	S	S	C	S
8.2	Rack type Automatic Cooling Bed	1 Set	S	S	S	S	S	C	S
8.3	Run Out Roller Table	1 No.	S	S	S	S	S	C	S
8.4	Cold Shear	1 No.	S	S	S	S	S	C	S
9	Continuous operating dividing shear for TMT Bars	1 No.	S	S	S	S	S	C	S
10	Bar Finishing & Bundling (Packing Bed)								
10.1	Cold Shear Run-Out Roller Table	1 No.	S	S	S	S	S	C	S
10.2	Packing Bed	1 No.	S	S	S	S	S	C	S
10.3	Roller Conveyor	1 No.	S	S	S	S	S	C	S
11	TMT SYSTEM	1 No.	S	S	S	S	S	C	S
16	UTM Machine	1 No.	S	S	S	S	S	C	S
	Bend / Rebind machine	1 No.	S	S	S	S	S	C	S



PREFERRED MAKE OF STANDARD BOUGHTOUT ITEMS

Main AC Motors	: Kirloskar Electrical Co. / Crompton Greaves
Small Motors for Auxiliary Equipment	: Kirloskar Electrical Co./ ABB / Siemens / Bharat Bijlee
D. C. Motors	: Kirloskar Electrical Co./ Integrated Electric Co.
D. C. Panels for Continuous Mill	: Parker / ABB
ACB, Switchgears, etc in AC Panels	: L&T / Siemens / ABB
Overload Relays	: Siemens / L&T
HRC Fuses	: L&T / Siemens
Starters	: Siemens/ ABB/Schneider/GE
Push Buttons	: Siemens / L&T / Telemechanique / IFM
PLC	: Siemens
Control / Selector Switch	: Kaycee / Jyothi / Telemechanique
Bearings	: SKF / FAG / TIMKEN / NACHI
MCBs	: Havells / L&T / Siemens
Meters (Ammeter, Voltmeter, P.F. Meters):	AE / Rishab / IMP
Multifunction Meters	: Konzerv / Trinity
Indicating Lamps	: Siemens / Tekmic / Telemechanique
Pumps	: Kirloskar Pumps Ltd. / KSB Make
M. S. Plates, M. S. Channel, Structure, Pipes, etc	: As per Indian Standard IS – 2062 & other Grade from Primary producer such as Jindal Steel / Essar Steel / Steel Authority of India
Hydraulic Equipment	: Rexroth / Parker / Yuken

All material will be of Standard Make & equivalent to 1st Class Maker



B. Consultation & Supervision of Erection, Installation & Commissioning of Rolling Mill**A. Scope of Services.**

1. Preparing Plant Lay-out
2. Detail Engineering of Mechanical, Electrical and civil work involved in the project.
3. To provide technical assistance & supervision for successful commissioning of the projects.
4. We shall depute a team of suitable persons for erection and Installation of the Steel Plant.
5. Supply of foundation plans with load data for all the equipment including requirement of embedded parts, trenches for cabling & piping etc. to enable customer arrange for design and construction of civil works.

B. Facilities to implement project.

The client will provide following -

1. All local manpower skilled and unskilled workers required for erection.
2. All civil work will be done by the client as per our drg. & instruction of our Engineer.
3. All tools, tackles and material handling equipment like cranes etc. raw materials, consumables, Oil & Lubricants, Water, compressed air, power etc. will be provided by the buyer at buyer's cost to facilitate erection & commissioning.
4. All expenses Air Ticket, visa charges, insurance, lodging, Boarding, local conveyance etc. shall be borne by the client.
5. You will cover all our personnel engaged at project site by suitable insurance policy according to status of the person.
6. You have to pay actual Salary of all Persons coming on Site for Erection & Installation
7. You have to provide all Staff for Hot trial & Production of the Plant.



EXCLUSIONS

1. Shed (Hanger), Cranes, Rails & DSL for Crane
2. HT Side, Transformers, all Cables, Capacitors & Earthing
3. All Civil Work and Mill Foundations including water & electrical trenches.
4. Fabrication on Site like –Trough, Walk ways, etc.
5. Re Heating Furnace, if required
6. Workshop Machinery including CNC Roll Branding & Rib Cutting machine
7. Interconnecting Air, Oil, Hydraulic & Water Pipe Line
8. Erection and Installation
9. First Fill of Lubricants

